

# groqClient.ts

Frontend | TypeScript

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/**
 * groqClient.ts
 *
 * Thin wrapper around the Groq SDK that adds:
 * - State-aware request construction
 * - Failure memory blacklist injection
 * - 1-second timeout with offline fallback
 * - Response parsing and intervention type detection
 */

import Groq from 'groq-sdk';
import { BIOSYNC_SYSTEM_PROMPT, buildUserMessage } from '../systemPrompt';
import { InterventionType } from '../hooks/useStateMemory';
import { CognitiveState } from '../utils/stressEngine';

// ■■■ Types ■■■
export interface AIRequest {
  stressScore: number;
  state: CognitiveState;
  failureMemory: string[]; // Intervention names that FAILED
  userInput?: string; // Transcribed speech / typed text
  sessionMinutes: number;
}

export interface AIResponse {
  text: string;
  interventionType: InterventionType;
  isOfflineFallback: boolean;
}

// ■■■ Offline fallback tree ■■■
// Used when Groq is unavailable or times out (e.g., Wi-Fi off demo moment)

const OFFLINE_RESPONSES: Record<CognitiveState, Array<{ text: string; type: InterventionType }>> = {
  CALM: [
    { text: "Cognitive bandwidth optimal. This is your deep work window.", type: 'DEEP_WORK_PROMPT' },
    { text: "Energy available. Lock in on one high-leverage task now.", type: 'FOCUS_REFRAME' },
  ],
  LOAD: [
    { text: "Write down the three things that matter most right now.", type: 'TASK_STRIPPING' },
    { text: "Close every tab not related to the current task.", type: 'FOCUS_REFRAME' },
    { text: "Set a 25-minute sprint. Nothing else exists until it ends.", type: 'MICRO_BREAK' },
  ],
  CRITICAL: [
    { text: "Stand up. Walk away from the screen for 60 seconds.", type: 'MICRO_BREAK' },
    { text: "Write your single most important next action on paper.", type: 'TASK_STRIPPING' },
    { text: "Four counts in, hold four, four counts out. Do it twice.", type: 'BREATHING_EXERCISE' },
    { text: "Change your environment. Different room, different light.", type: 'ENVIRONMENTAL_RESET' },
    { text: "Notify a teammate. Step away for 10 minutes.", type: 'ESCALATE_HUMAN' },
  ],
};

// ■■■ Client initialization ■■■

const groq = new Groq({
  apiKey: process.env.NEXT_PUBLIC_GROQ_API_KEY ?? '',
  dangerouslyAllowBrowser: true,
});

// ■■■ Main function ■■■

/**
 * Fetches a state-aware cognitive intervention from Groq Llama 3.1.
 * Falls back to deterministic offline responses on timeout or network failure.
 *
 * @param request - Stress context including score, state, and failure memory

```

